

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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1. TITLE

Design a Static Webpage Using HTML Components

2. AIM

To design and develop a static webpage using various HTML components such as headings, lists, links, images, tables and forms, without using CSS or JavaScript.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
- To use heading tags to organize content hierarchically.
- To create ordered and unordered lists.
- To embed images using the tag with alt attributes.
- To design a table to display structured data.
- To create a form to collect user input.
- To render the completed page in a browser and verify all components.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages.

An HTML document is built using nested elements, each represented by opening and closing tags.

Heading tags (<h1> to <h6>) are used to define headings of different levels, with <h1> being the most important.

Lists can be unordered () with bullet points, or ordered () with numbering.

The tag embeds images into a page and requires a src attribute for the image source and an alt attribute for accessibility.

The <table> tag, along with <tr>, <th> and <td>, is used to organize data into rows and columns.

The <form> tag, along with input elements such as <input>, <select> and <textarea>, is used to collect data from users.

Even without CSS for styling, standard browser defaults render HTML elements in a clean and readable manner.

5. ALGORITHM / PROCEDURE

- Create a new project folder for the experiment.
- Create a new file named index.html.
- Add the basic HTML boilerplate (<!DOCTYPE html>, <html>, <head>, <body>).
- Add a page title within the <title> tag.
- Insert heading tags to define the page title and section headings.
- Insert an unordered list containing navigation links.
- Add a horizontal rule to separate sections.
- Insert a paragraph describing the purpose of the page.
- Insert a nested list describing key HTML elements used.
- Embed an image using the tag with an appropriate alt attribute.
- Create a table with a header row and multiple data rows.
- Create a form with text input, email input, a dropdown and a textarea.
- Add submit and reset buttons to the form.
- Save the file and open it in a web browser to verify the output.
- Test all links, table rows and form fields for correct rendering.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>My Static HTML Page</title>
</head>
<body>

    <h1>My Static HTML Page</h1>
    <h2>Built completely from scratch</h2>

    <ul>
        <li><a href="#about">About This Page</a></li>
        <li><a href="#features">Feature Matrix</a></li>
        <li><a href="#contact">Get In Touch</a></li>
        <li><a href="https://en.wikipedia.org/wiki/HTML" target="_blank">Visit Wikipedia
(External Link)</a></li>
    </ul>

    <hr>

    <h2 id="about">About This Page</h2>
    <p>HTML (HyperText Markup Language) forms the skeleton of every website...</p>

    <h3>Key HTML Elements Used Here:</h3>
    <ul>
```

```

    <li><strong>Headings:</strong> Organized structurally from <h1> down to
    <h3>.</li>
    <li><strong>Lists:</strong> Ordered (numbered) and unordered (bulleted) lists.</li>
    <li><strong>Media:</strong> Embedded images with alternative descriptive text.</li>
  </ul>

  
  <p><em>Figure 1: A placeholder image loaded directly via URL.</em></p>

  <hr>

  <h2 id="features">HTML Component Table</h2>
  <p>Tables are great for displaying structured data sets.</p>

  <table border="1" cellpadding="6" cellspacing="0">
    <thead>
      <tr>
        <th>Tag Name</th>
        <th>Primary Function</th>
        <th>Type</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td><code>&lt;a&gt;</code></td>
        <td>Creates hyperlinks to other URLs or page sections.</td>
        <td>Inline</td>
      </tr>
      <tr>
        <td><code>&lt;table&gt;</code></td>
        <td>Organizes data into rows and columns.</td>
        <td>Block</td>
      </tr>
      <tr>
        <td><code>&lt;form&gt;</code></td>
        <td>Collects input data from users.</td>
        <td>Block</td>
      </tr>
    </tbody>
  </table>

  <hr>

  <h2 id="contact">Contact Form</h2>
  <p>Forms let users input data, though a backend server or service is required to process
  it.</p>

  <form action="#" method="POST">
    <label for="fullname">Full Name:</label><br>
    <input type="text" id="fullname" name="fullname" placeholder="John Doe">
    <br><br>

    <label for="email">Email Address:</label><br>
    <input type="email" id="email" name="email" placeholder="john@example.com">
    <br><br>

    <label for="reason">Reason for Contact:</label><br>
    <select id="reason" name="reason">
      <option value="general">General Inquiry</option>
      <option value="support">Support</option>

```

```

        <option value="feedback">Feedback</option>
    </select>
    <br><br>

    <label for="message">Your Message:</label><br>
    <textarea id="message" name="message" rows="4" cols="40" placeholder="Type your
message here..."></textarea>
    <br><br>

    <button type="submit">Submit Form</button>
    <button type="reset">Clear Form</button>
</form>

<hr>

<footer>
    <p>&copy; 2026 My Static HTML Page. Built completely from scratch.</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<h1> to <h3>	Define headings of decreasing importance to structure the page.
/	Creates an unordered (bulleted) list of items.
	Creates a hyperlink to another section or an external page.
	Embeds an image and provides alternative text for accessibility.
<table>, <tr>, <th>, <td>	Organizes tabular data into rows, header cells and data cells.
<form action="..." method="POST">	Defines a form that collects and submits user input.
<input>, <select>, <textarea>	Collect different types of user input within the form.
<button type="submit">	Submits the form data when clicked.

8. TEST CASES AND EXECUTION DATA

The static page was verified against the following test cases and results were recorded after execution.

Test Case	Component Tested	Expected Result	Actual Result	Status
TC-01	Heading tags	Headings displayed in decreasing font size (h1 > h2 > h3)	Headings displayed correctly	Pass
TC-02	Navigation list links	Clicking a link scrolls to the corresponding section	Navigation worked as expected	Pass
TC-03	Image with alt text	Image displayed; alt text shown if image fails to load	Image and alt text rendered correctly	Pass

TC-04	HTML component table	Table displayed with header row and 3 data rows	Table displayed correctly	Pass
TC-05	Contact form fields	Text, email, dropdown and textarea fields accept input	All fields accepted input correctly	Pass
TC-06	Submit / Clear buttons	Clear button resets all form fields	Form reset successfully	Pass

9. OUTPUT

Output 1: Page Header, Navigation and About Section

My Static HTML Page

Built completely from scratch

- [About This Page](#)
- [Feature Matrix](#)
- [Get In Touch](#)
- [Visit Wikipedia \(External Link\)](#)

About This Page

HTML (HyperText Markup Language) forms the skeleton of every website on the internet. Even without CSS for styling or JavaScript for logic, standard browser defaults render HTML elements cleanly and accessibly.

Key HTML Elements Used Here:

- **Headings:** Organized structurally from <h1> down to <h3>.
- **Lists:** Ordered (numbered) and unordered (bulleted) lists.
- **Media:** Embedded images with alternative descriptive text.



Figure: Headings, navigation list and About section

Output 2: Image, HTML Component Table



Figure 1: A placeholder image loaded directly via URL.

HTML Component Table

Tables are great for displaying structured data sets.

Tag Name	Primary Function	Type
<a>	Creates hyperlinks to other URLs or page sections.	Inline
<table>	Organizes data into rows and columns.	Block
<form>	Collects input data from users.	Block

Contact Form

Figure: Embedded image and HTML component table

Output 3: Contact Form

Contact Form

Forms let users input data, though a backend server or service is required to process it.

Full Name:

Email Address:

Reason for Contact:

Your Message:

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Figure: Contact form with input fields and buttons

10. RESULT

Thus, a static webpage was successfully designed using various HTML components such as headings, lists, links, images, a table and a form. All components were rendered correctly in the browser and verified using multiple test cases.